

Towards Minimal Axioms for Probabilistic Programming

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What does this monad need to be a good model of probability?

Related question: what is special about the *Giry monad* $\mathbf{G} : \mathbf{Meas} \rightarrow \mathbf{Meas}$ on measurable spaces, where $\mathbf{G}X$ is the space of bona fide probability measures on X ?

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We restrict to the class of *standard Borel* measurable spaces and consider instead the restricted $\mathbf{G} : \mathbf{Borel} \rightarrow \mathbf{Borel}$.

Theorem

$G : \mathbf{Borel} \rightarrow \mathbf{Borel}$ is initial among affine commutative monads which preserve countable cofiltered limits, pullbacks along coproduct inclusions, and contain a commutative binary operation $\mathbf{cf} : G(2)$.

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Let's consider these properties in turn. First, $\mathbf{cf}$ is just the uniform distribution on two points.

G preserves pullbacks along coproduct inclusions: Given $f : X \rightarrow A + B$, a distribution on $f^{-1}(A)$ is equivalently a distribution on X so that the image is concentrated on A .

$$\begin{array}{ccc} G(f^{-1}(A)) & \longrightarrow & G(X) \\ \downarrow & & \downarrow f \\ G(A) & \longrightarrow & G(A + B) \end{array}$$

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In a dependently typed language, we can internalize the limits involved in this theorem, and add operators implementing the required isomorphism.

$$\text{com} : \sum_{p : \prod_{n:\mathbb{N}} P(X^n)} \prod_{n:\mathbb{N}} P(\pi_{1,\dots,n-1})(p_n) =_{P(X^{n-1})} p(n-1) \rightarrow P(X^{\mathbb{N}})$$

(We could also replace $X^{\mathbb{N}}$ with a dependent function type here, of course).

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$$\text{com} : \sum_{p : \prod_{n:\mathbb{N}} P(X^n)} \prod_{n:\mathbb{N}} P(\pi_{1,\dots,n-1})(pn) =_{P(X^{n-1})} p(n-1) \rightarrow P(X^{\mathbb{N}})$$

(We could also replace $X^{\mathbb{N}}$ with a dependent function type here, of course).

The question is

- Does this allow us to express useful probabilistic programs?
- Can we prove a reasonable number of equations between them?

Theorem

Let $\mathcal{C}$ be an elementary topos with countable choice, and let $P : \mathbf{C} \rightarrow \mathbf{C}$ be an affine commutative monad which preserves countable cofiltered limits and pullbacks along all monomorphisms, and which has a commutative binary operation. Then there exists a map $b : [0, 1] \rightarrow P(2)$ so that the induced $[0, 1] \times P(X)^2 \rightarrow P(X)$ makes every $P(X)$ into an internal convex space.

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(This is work in progress) In other words, we can define the Bernoulli distribution $b(\lambda) \in P(2)$ for each $\lambda \in [0, 1]$, and the expected equations between finitary distributions defined in terms of these hold. (This may hold without countable choice, with the Cauchy reals.)

Conclusion

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Ongoing questions:

- Under what assumption on $\mathcal{C}$ can we prove which identities between distributions using only these assumptions on P ?

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Ongoing questions:

- Under what assumption on $\mathbf{C}$ can we prove which identities between distributions using only these assumptions on P ?
- Can we describe the initial monad in a useful way?
- Is there a useful version of this without dependent types?

Thank you for listening.

Paper: “The Universal Property of Measure-Theoretic Probability”, [arxiv:2512.15485](https://arxiv.org/abs/2512.15485)